

Supplementary Information

Status of a decennial marine invasion by the bisexual mussel *Semimytilus algosus* (Gould, 1850) in South Africa

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Supplementary Table S1: Additional records of the alien species *Semimytilus algosus* in Namibia

Year	Site	Latitude (°S)	Longitude (°E)	Date	Occurrence	Source
1990	Lüderitz	26.708400	15.086000	—	Present, isolated individuals	Bustamante (1994)
2004–2009	Green Precipice	27.789301	15.583088	—	Not detected	Namdeb database
2004–2013	Wolf Bay	26.807601	15.124260	—	Not detected	Namdeb database
	Atlas Bay	26.829047	15.139073	—	Not detected	Namdeb database
	East Whale	26.907868	15.162987	—	Not detected	Namdeb database
	Elizabeth Bay Point, exposed	26.924150	15.191883	—	Not detected	Namdeb database
	Elizabeth Bay Point, semi-exposed	26.922533	15.192250	—	Not detected	Namdeb database
	South Jetty	26.920350	15.192017	—	Not detected	Namdeb database
	North Jetty	26.918383	15.191567	—	Not detected	Namdeb database
	Elizabeth Bay South	26.935617	15.230100	—	Not detected	Namdeb database
	Bogenfels	27.457255	15.390880	—	Not detected	Namdeb database
	Cabin	27.607308	15.496700	—	Not detected	Namdeb database
	Dernberg Bay	27.710400	15.531070	—	Not detected	Namdeb database
	Site 1	27.855287	15.647800	—	Not detected	Namdeb database
	Panther Reef	27.915260	15.684800	—	Not detected	Namdeb database
	West Whale	26.908933	15.160750	—	Not detected	Namdeb database
2007–2013	Hostel (H155-160)	28.047183	15.765467	—	Not detected	Pulfrich (2013)
	No. 1 Plant (H35-50)	28.127100	15.849950	—	Not detected	Pulfrich (2013)
	Kerbe Huk (K65-69)	28.231317	15.983483	—	Not detected	Pulfrich (2013)
	No. 2 Plant (M170-175)	28.279517	16.045917	—	Not detected	Pulfrich (2013)

2014	Mittag (M80-85)	28.338967	16.106617	—	Not detected	Pulfrich (2013)
	Wolf Bay	26.807601	15.124260	—	Not detected	Namdeb database
	Atlas Bay	26.829047	15.139073	—	Not detected	Namdeb database
	West Whale	26.908933	15.160750	—	Not detected	Namdeb database
	East Whale	26.907868	15.162987	—	Not detected	Namdeb database
	Elizabeth Bay Point, exposed	26.924150	15.191883	—	Not detected	Namdeb database
	Elizabeth Bay Point, semi-exposed	26.922533	15.192250	—	Not detected	Namdeb database
	South Jetty	26.920350	15.192017	—	Not detected	Namdeb database
	North Jetty	26.918383	15.191567	—	Not detected	Namdeb database
	Elizabeth Bay South	26.935617	15.230100	—	Not detected, sand inundation	Namdeb database
	Hostel (H155-160)	28.047183	15.765467	—	Not detected	Pulfrich (2014)
	No. 1 Plant (H35-50)	28.127100	15.849950	—	Not detected	Pulfrich (2014)
	Kerbe Huk (K65-69)	28.231317	15.983483	—	Not detected	Pulfrich (2014)
	No. 2 Plant (M170-175)	28.279517	16.045917	—	Not detected	Pulfrich (2014)
2015	Mittag (M80-85)	28.338967	16.106617	November	Present, few individuals	Pulfrich (2014)
	Wolf Bay	26.807601	15.124260	—	Not detected	Namdeb database
	Atlas Bay	26.829047	15.139073	—	Not detected	Namdeb database
	West Whale	26.908933	15.160750	—	Not detected	Namdeb database
	East Whale	26.907868	15.162987	—	Not detected	Namdeb database
	Elizabeth Bay Point, exposed	26.924150	15.191883	—	Not detected	Namdeb database
	Elizabeth Bay Point, semi-exposed	26.922533	15.192250	—	Not detected	Namdeb database
	North Jetty	26.918383	15.191567	—	Not detected	Namdeb database
	South Jetty	26.920350	15.192017	—	Not detected	Namdeb database
	Elizabeth Bay South	26.935617	15.230100	—	Not detected, sand inundation	Namdeb database
	Hostel (H155-160)	28.047183	15.765467	—	Not detected	Pulfrich (2015)
	No. 1 Plant (H35-50)	28.127100	15.849950	—	Present, mean of 0.8% cover	Pulfrich (2015)
	Kerbe Huk (K65-69)	28.231317	15.983483	—	Not detected	Pulfrich (2015)
	No. 2 Plant (M170-175)	28.279517	16.045917	—	Present, mean of <0.1% cover	Pulfrich (2015)
2016	Mittag (M80-85)	28.338967	16.106617	—	Present, mean of 11.1% cover	Pulfrich (2015)
	Wolf Bay	26.807601	15.124260	—	Not detected	Namdeb database
	Atlas Bay	26.829047	15.139073	—	Not detected	Namdeb database
	West Whale	26.908933	15.160750	—	Not detected	Namdeb database
	East Whale	26.907868	15.162987	—	Not detected	Namdeb database
	Elizabeth Bay Point, exposed	26.924150	15.191883	—	Not detected	Namdeb database

2017	Elizabeth Bay Point, semi-exposed	26.922533	15.192250	—	Not detected	Namdeb database
	South Jetty	26.920350	15.192017	—	Not detected	Namdeb database
	North Jetty	26.918383	15.191567	—	Not detected	Namdeb database
	Elizabeth Bay South	26.935617	15.230100	—	Not detected, sand inundation	Namdeb database
	Hostel (H155-160)	28.047183	15.765467	—	Not detected	Pulfrich (2016)
	No. 1 Plant (H35-50)	28.127100	15.849950	—	Not detected	Pulfrich (2016)
	Kerbe Huk (K65-69)	28.231317	15.983483	—	Present, mean of 10.1% cover	Pulfrich (2016)
	No. 2 Plant (M170-175)	28.279517	16.045917	—	Present, mean of 5.1% cover	Pulfrich (2016)
	Mittag (M80-85)	28.338967	16.106617	—	Not detected	Pulfrich (2016)
	Wolf Bay	26.807601	15.124260	—	Not detected	Namdeb database
	Atlas Bay	26.829047	15.139073	—	Not detected	Namdeb database
	West Whale	26.908933	15.160750	—	Not detected	Namdeb database
	East Whale	26.907868	15.162987	—	Not detected	Namdeb database
	Elizabeth Bay Point, exposed	26.924150	15.191883	—	Not detected	Namdeb database
	Elizabeth Bay Point, semi-exposed	26.922533	15.192250	—	Not detected	Namdeb database
	South New Jetty	26.921200	15.192450	—	Not detected	Namdeb database
2018	South Jetty	26.920350	15.192017	—	Not detected	Namdeb database
	North Jetty	26.918383	15.191567	—	Present, mean of 0.2% cover	Namdeb database
	Elizabeth Bay South	26.935617	15.230100	—	Not detected, sand inundation	Namdeb database
	Hostel (H155-160)	28.047183	15.765467	—	Present, mean of <0.1% cover	Pulfrich (2017)
	No. 1 Plant (H35-50)	28.127100	15.849950	—	Not detected	Pulfrich (2017)
	Kerbe Huk (K65-69)	28.231317	15.983483	—	Present, mean of 18.6% cover	Pulfrich (2017)
	No. 2 Plant (M170-175)	28.279517	16.045917	—	Present, mean of 8.6% cover	Pulfrich (2017)
	Mittag (M80-85)	28.338967	16.106617	—	Present, mean of 6.7% cover	Pulfrich (2017)
	Wolf Bay	26.807601	15.124260	—	Not detected	Namdeb database
	Atlas Bay	26.829047	15.139073	—	Not detected	Namdeb database
	West Whale	26.908933	15.160750	—	Present, mean of 0.3% cover	Namdeb database
	East Whale	26.907868	15.162987	—	Not detected	Namdeb database
	Elizabeth Bay Point, exposed	26.924150	15.191883	—	Not detected	Namdeb database
	Elizabeth Bay Point, semi-exposed	26.922533	15.192250	—	Present, mean of <0.1% cover	Namdeb database
	South New Jetty	26.921200	15.192450	—	Present, mean of <0.1% cover	Namdeb database
	South Jetty	26.920350	15.192017	—	Present, mean of <0.1% cover	Namdeb database
	North Jetty	26.918383	15.191567	—	Not detected, sand inundation	Namdeb database
	Elizabeth Bay South	26.935617	15.230100	—	Not detected, sand inundation	Namdeb database

2019	Hostel (H155-160)	28.047183	15.765467	—	Present, mean of 0.1% cover	Pulfrich (2018)
	No. 1 Plant (H35-50)	28.127100	15.849950	—	Present, mean of 0.3% cover	Pulfrich (2018)
	Kerbe Huk (K65-69)	28.231317	15.983483	—	Present, mean of 1.7% cover	Pulfrich (2018)
	No. 2 Plant (M170-175)	28.279517	16.045917	—	Present, mean of 5.9% cover	Pulfrich (2018)
	Mittag (M80-85)	28.338967	16.106617	—	Present, mean of 9.7% cover	Pulfrich (2018)
	Portuguese Lorry North	17.564700	11.736670	27 September	Not detected	Kreiner et al. (2019a)
	Portuguese Lorry South	17.585444	11.736253	27 September	Not detected	Kreiner et al. (2019a)
	Angra Fria	18.285722	11.962917	28 September	Not detected	Kreiner et al. (2019a)
	False Cape Fria North	18.480803	12.023750	28 September	Present, up to 68.4% cover	Kreiner et al. (2019a)
	False Cape Fria South	18.484378	12.025975	28 September	Present, up to 70.2% cover	Kreiner et al. (2019a)
	Rocky Point, sheltered	18.994167	12.476361	29 September	Not detected	Kreiner et al. (2019a)
	Rocky Point Bench 1, exposed	18.994614	12.475358	29 September	Present, up to 4.6% cover	Kreiner et al. (2019a)
	Rocky Point Bench 2, exposed	18.994744	12.475633	29 September	Present, up to 15.8% cover	Kreiner et al. (2019a)
	North of Möwe Bay, sheltered	19.348181	12.706558	30 September	Not detected	Kreiner et al. (2019a)
	Möwe Bay Bench 1, semi-exposed	19.381747	12.704403	1 October	Present, up to 0.2% cover	Kreiner et al. (2019a)
	Möwe Bay Bench 2, exposed	19.381756	12.704403	1 October	Present, up to 10.8% cover	Kreiner et al. (2019a)
	Students Bay Bench 1	20.133561	13.125803	2 October	Present, up to 97.4% cover	Kreiner et al. (2019a)
	Students Bay Bench 2	20.133575	13.125778	2 October	Present, up to 79.4% cover	Kreiner et al. (2019a)
	South of Möwe Bay, seal colony	20.133828	13.117783	1 October	Present, up to 1.2% cover	Kreiner et al. (2019a)
	Henties Bay, Solitude South	22.150833	14.288056	4 July	Present, up to 7.1% cover	Kreiner et al. (2019b)
	Mile Four	22.618972	14.520861	3 July	Not detected	Kreiner et al. (2019b)
	Patrysberg	22.741083	14.524333	5 July	Present, up to 6% cover	Kreiner et al. (2019b)
	Long Beach, Swakopmund	22.817306	14.541833	2 July	Present, up to 67.4% cover	Kreiner et al. (2019b)
	Meob	24.536667	14.598611	3 August	Present, up to 92% cover	Kreiner et al. (2019c)
	Spencer Bay North	25.621944	14.848611	1 August	Present, up to 4.2% cover	Kreiner et al. (2019c)
	Spencer Bay South	25.734444	14.847778	1 August	Present, up to 25.6% cover	Kreiner et al. (2019c)
	Diaz Point, sheltered	26.643333	15.090556	31 July	Not detected	Kreiner et al. (2019c)
	Diaz Point, semi-exposed	26.654722	15.082500	31 July	Not detected	Kreiner et al. (2019c)
	Grossebucht, semi-exposed	26.739722	15.090556	30 July	Not detected	Kreiner et al. (2019c)
	Grossebucht, sheltered	26.738333	15.094444	30 July	Not detected	Kreiner et al. (2019c)
	Wolf Bay	26.807601	15.124260	—	Not detected	Namdeb database
	Atlas Bay	26.829047	15.139073	—	Not detected	Namdeb database
	West Whale	26.908933	15.160750	—	Present, mean of <0.1% cover	Namdeb database
	East Whale	26.907868	15.162987	—	Not detected	Namdeb database

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South Jetty	26.920350	15.192017	—	Not detected	Namdeb database
North Jetty	26.918383	15.191567	—	Not detected, sand inundation	Namdeb database
Elizabeth Bay South	26.935617	15.230100	—	Not detected, sand inundation	Namdeb database

References

- Bustamante RH. 1994. Patterns and causes of intertidal community structure around the coast of southern Africa. PhD thesis, University of Cape Town, South Africa.
- Kreiner A, Branch G, Ferreira L, Reddy M, Serrão E, Thomas M. 2019a. Report: Coastal Biodiversity Surveys – Survey 3. Subdivision Environment, National Marine Information and Research Centre, Ministry of Fisheries and Marine Resources, Namibia.
- Kreiner A, Bolton J, Branch G, Lima F, Reddy M, Serrão E, Thomas M. 2019b. Report: Coastal Biodiversity Surveys – Survey 1. Subdivision Environment, National Marine Information and Research Centre, Ministry of Fisheries and Marine Resources, Namibia.
- Kreiner A, Reddy M, Seabra R, Thomas M. 2019c. Report: Coastal Biodiversity Surveys – Survey 2. Subdivision Environment, National Marine Information and Research Centre, Ministry of Fisheries and Marine Resources, Namibia.
- Namdeb database [NAMDEB Diamond Corporation (Pty) Ltd]. Long-term coastal biological monitoring programme database [accessed 30 May 2020].
- Pulfrich A. 2013. Intertidal rocky-shore communities of the Sperrgebiet coastline: Consolidated Rocky-shores Monitoring Report – 2013. Report prepared for NAMDEB Diamond Corporation (Pty) Ltd, Oranjemund, Namibia.
- Pulfrich A. 2014. Intertidal rocky-shore communities of the Sperrgebiet coastline: Consolidated Rocky-shores Monitoring Report – 2014. Report prepared for NAMDEB Diamond Corporation (Pty) Ltd, Oranjemund, Namibia.
- Pulfrich A. 2015. Intertidal rocky-shore communities of the Sperrgebiet coastline: Consolidated Rocky-shores Monitoring Report – 2015. Report prepared for NAMDEB Diamond Corporation (Pty) Ltd, Oranjemund, Namibia.
- Pulfrich A. 2016. Intertidal rocky-shore communities of the Sperrgebiet coastline: Consolidated Rocky-shores Monitoring Report – 2016. Report prepared for NAMDEB Diamond Corporation (Pty) Ltd, Oranjemund, Namibia.
- Pulfrich A. 2017. Intertidal rocky-shore communities of the Sperrgebiet coastline: Consolidated Rocky-shores Monitoring Report – 2017. Report prepared for NAMDEB Diamond Corporation (Pty) Ltd, Oranjemund, Namibia.
- Pulfrich A. 2018. Intertidal rocky-shore communities of the Sperrgebiet coastline: Consolidated Rocky-shores Monitoring Report – 2018. Report prepared for NAMDEB Diamond Corporation (Pty) Ltd, Oranjemund, Namibia.